

I. Personal and study details

Student's name: **Martan Matyáš**

Personal ID number: **525903**

Faculty / Institute: **Faculty of Electrical Engineering**

Department / Institute: **Department of Measurement**

Study program: **Open Informatics**

Specialisation: **Internet of Things**

II. Bachelor's thesis details

Bachelor's thesis title in English:

LionKey (security key) NFC support

Bachelor's thesis title in Czech:

LionKey (bezpečnostní klíč) rozšíření NFC

Guidelines:

For the LionKey project: An open-source FIDO2 USB Security Key, implement basic support for NFC technology.

The work consists of the following parts:

- 1) Familiarization with the existing codebase and the technologies needed to implement NFC.
- 2) Implement basic NFC functionality on the STEVAL-ST25R3916B hardware.
- 3) Demonstrate basic functionality - e.g., authentication on <https://webauthn.io>

Bibliography / sources:

<https://github.com/pokusew/lionkey>

https://dspace.cvut.cz/bitstream/handle/10467/123886/F3-DP-2025-Endler-Martin-FIDO2_USB_Security_Key.pdf?sequence=1&isAllowed=